PIMA DIABETES

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AGENDA

Project Overview:

Data Cleaning

Feature Engineering

Visual Exploration

Modeling & Prediction

Conclusions & Future Work

PROJECT OVERVIEW

Objective: Predict diabetes likelihood (Outcome: Diabetic/Not Diabetic) in 768 Pima Indian female patients using 9 features (e.g., Glucose, BMI, Insulin).

Core Problem: "Impossible" zero values (e.g., Glucose=0) mask missing—biologically invalid, biasing models.

Dataset Snapshot: 65% non-diabetic (500 cases), 35% diabetic (268); skewed distributions (e.g., Insulin median=30.5 pre-cleaning).

Workflow: Clean anomalies → Engineer features → Visualize patterns → Model & evaluate (Logistic Regression + KNN).

Tools: R in Google Colab (version control via shared notebooks); outputs saved as CSVs/PDFs

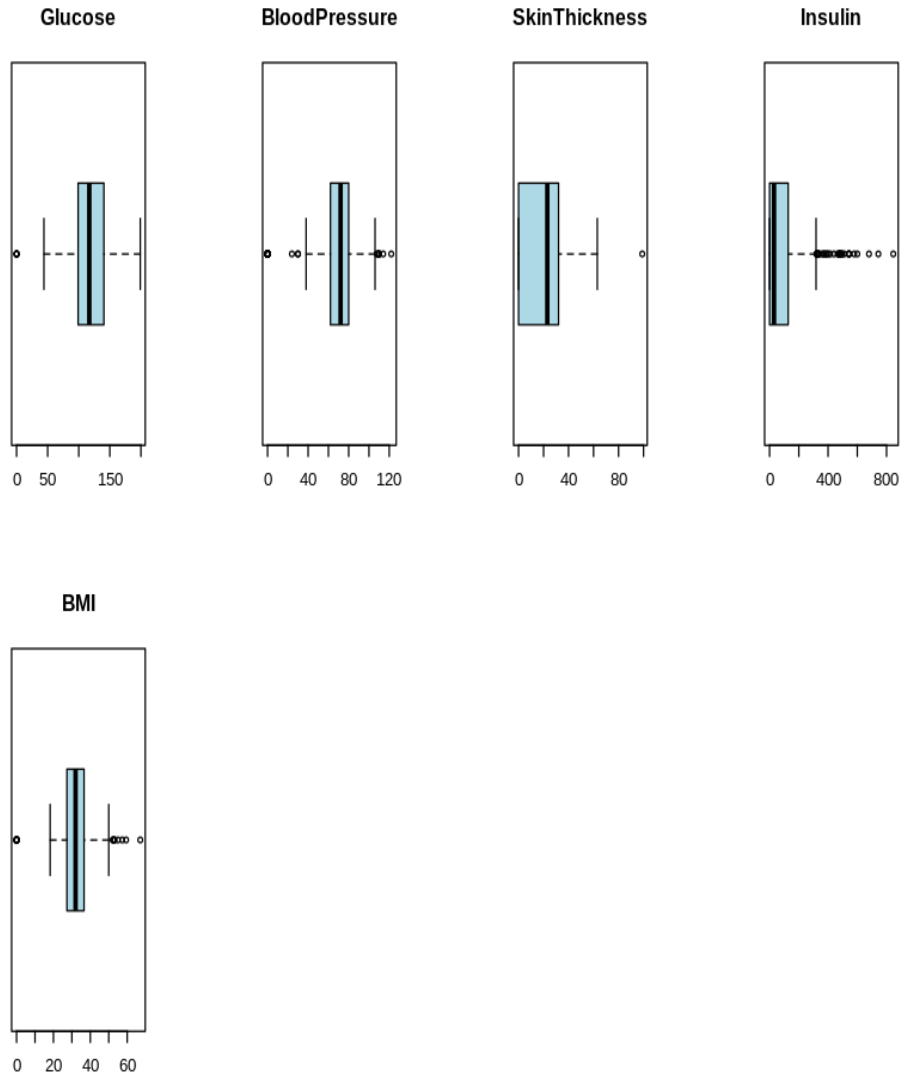
DATA CLEANING: IDENTIFYING ISSUES

Exploratory Analysis: Used summary() & str()—no negatives, but zeros in key columns: Glucose (5), BloodPressure (35), SkinThickness (227), Insulin (374), BMI (11).

Domain Knowledge: Zeros invalid (e.g., Glucose ≥44 mg/dL biologically); exempt Pregnancies/Age/Pedigree/Outcome.

Initial Replacement: Set zeros to NA—total NAs: Glucose=5, Insulin=374 (49% of rows affected).

Outlier Check: Boxplots revealed extremes (e.g., Insulin up to 846); histograms showed right-skew in Insulin/SkinThickness.



Data Cleaning: Imputation & Verification

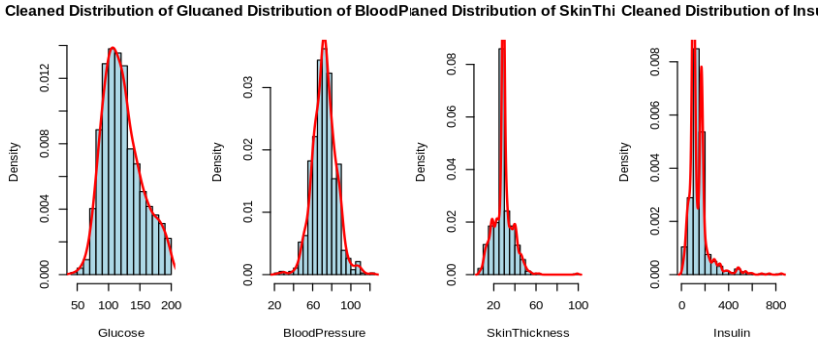
Method Choice: Median for robustness (skewed data); Insulin stratified by Outcome (non-diabetic: 102.5 μU/mL; diabetic: 169.5 μU/mL–plausible per medical norms).

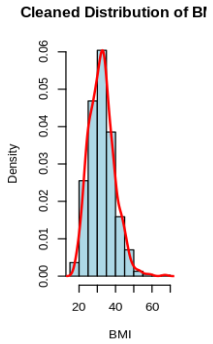
Other Medians: Glucose=117, BloodPressure=72, SkinThickness=29, BMI=32.3.

Post-Imputation Stats: No NAs/zeros; Insulin mean=141.8 (up from 79.8); BMI mean=32.46.

Verification: Histograms confirm improved distributions (e.g., Insulin less zero-inflated).

FEATURE ENGINEERING: STANDARDIZATION





Rationale:
Features on disparate scales (e.g., Age 21-81 vs. Pedigree 0-2.42)—standardization (z-score: mean=0, SD=1) aids distance-based models like KNN.

Applied to:
8 numerics (Pregnancies, Glucose, etc.)—scale() function.

Results:
All means ≈0, SDs=1 (e.g., Glucose mean=-4.68e-18

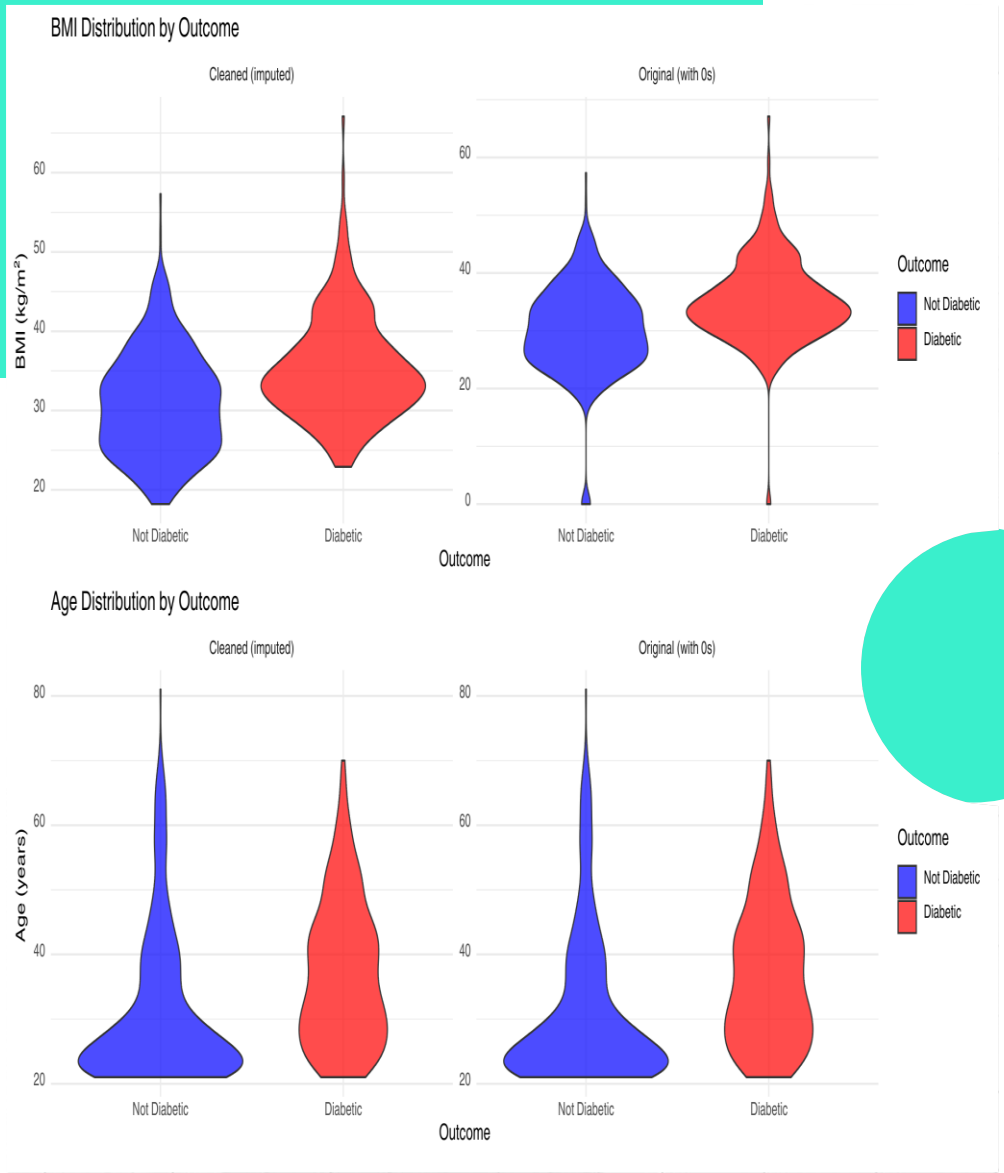
Saved:

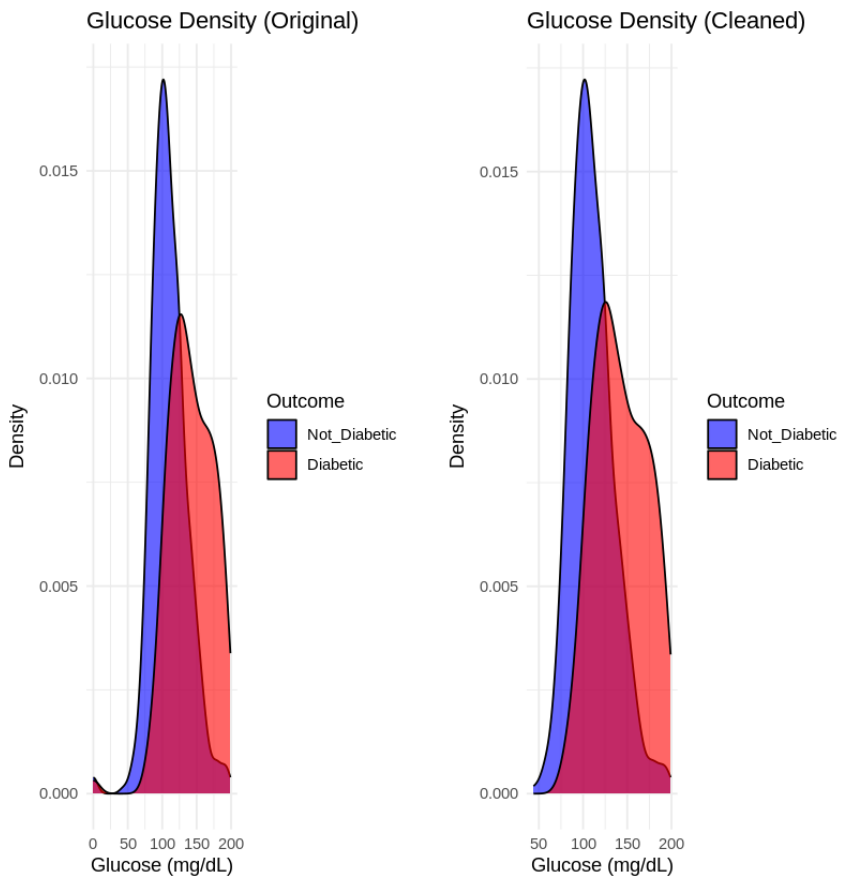
enhanced_diabetes.csv (no data loss).

FEATURE ENGINEERING: CATEGORIZATION

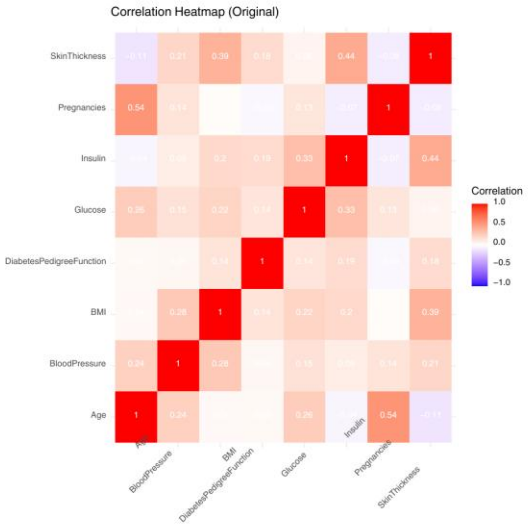
BMI Categories (WHO Thresholds)	Blood Pressure Categories	Target Refinement
Underweight(<18.5: 4 cases)	Normal (<80:563 cases)	"Not Diabetic" (500)
Normal (18.5-24.9: 101)	Elevated (80-89: 145)	"Diabetic" (268)
Overweight (25-29.9: 175)	High (≥90: 60)—diastolic focus.	
Obese (≥30: 488)		

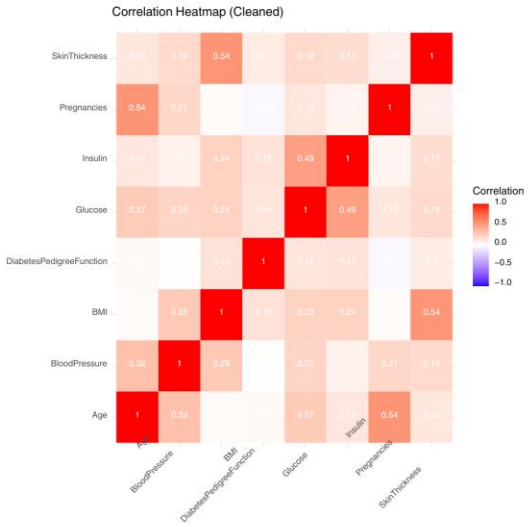
VISUAL EXPLORATION: GLUCOSE & BMI/AGE DISTRIBUTIONS

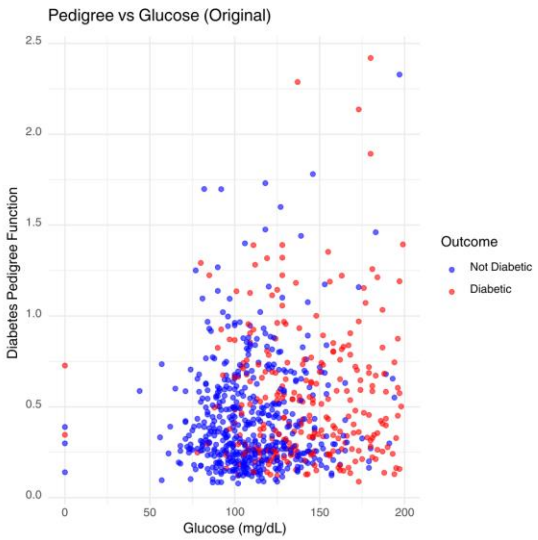


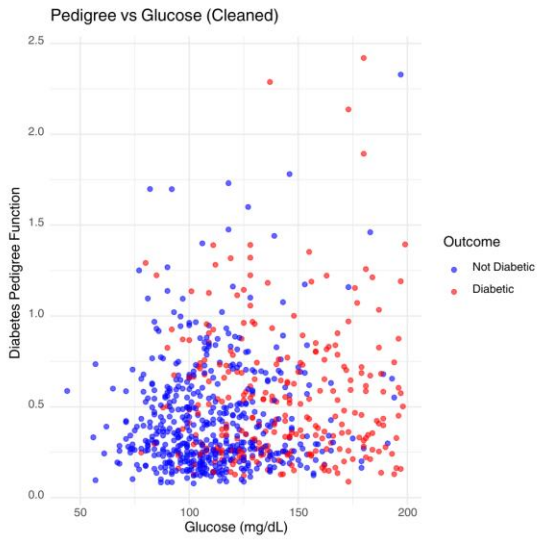


VISUALIZATION: CORRELATIONS & PEDIGREE



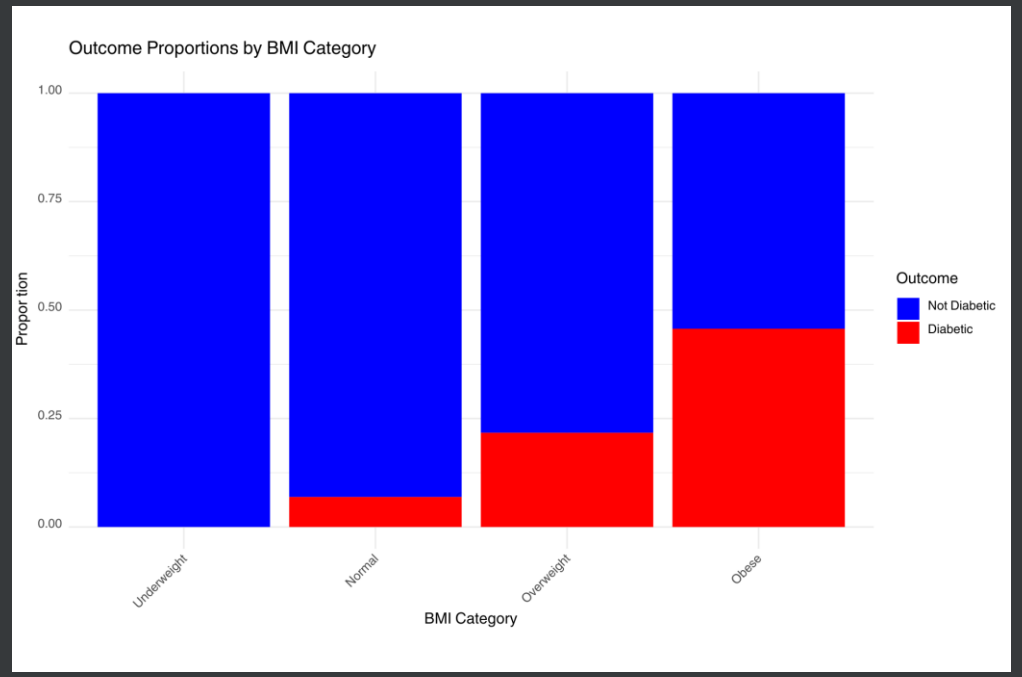


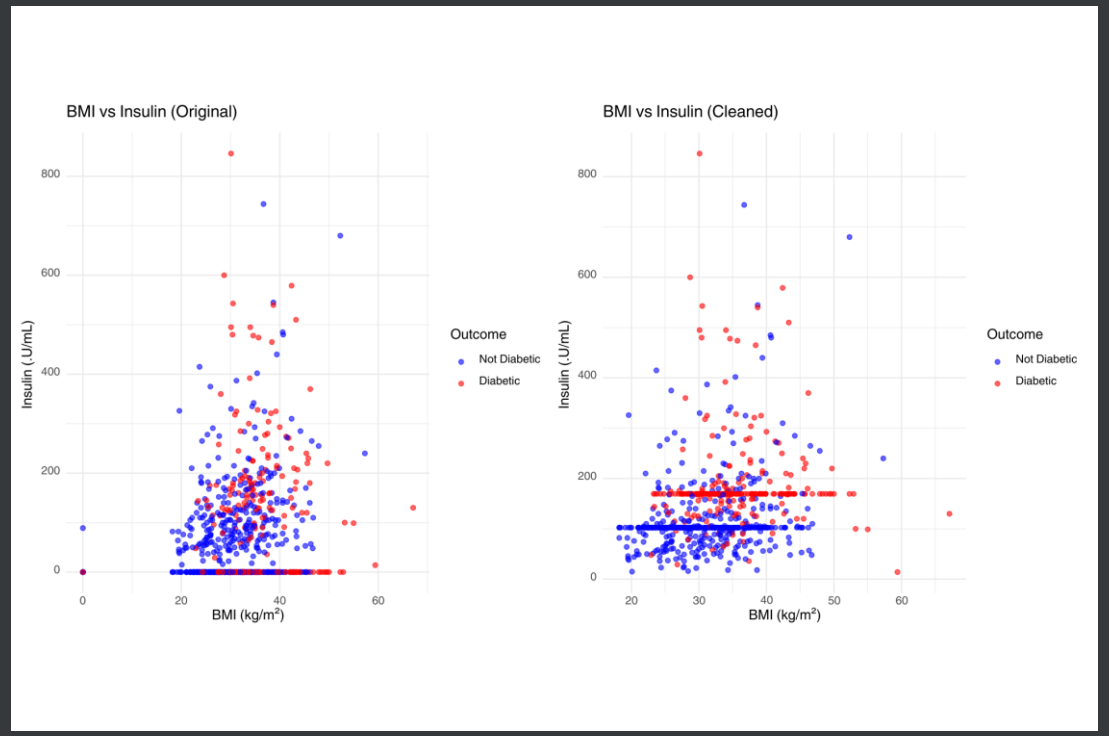




Visualization: Additional Insights

Key Takeaway: Vizes confirm obesity/glucose as modifiable risks; 2-3% better separation post-cleaning.





MODELING: TRAINING & PERFORMANCE

Setup: 10-fold CV (caret::train); standardized data; metrics: AUC, Accuracy, Sens/Spec.

Logistic Regression: Interpretable coeffs (e.g., Glucose +0.92 strongest positive); AUC=0.864, Accuracy=78.4% (Sens=87.4%, Spec=57.1%)

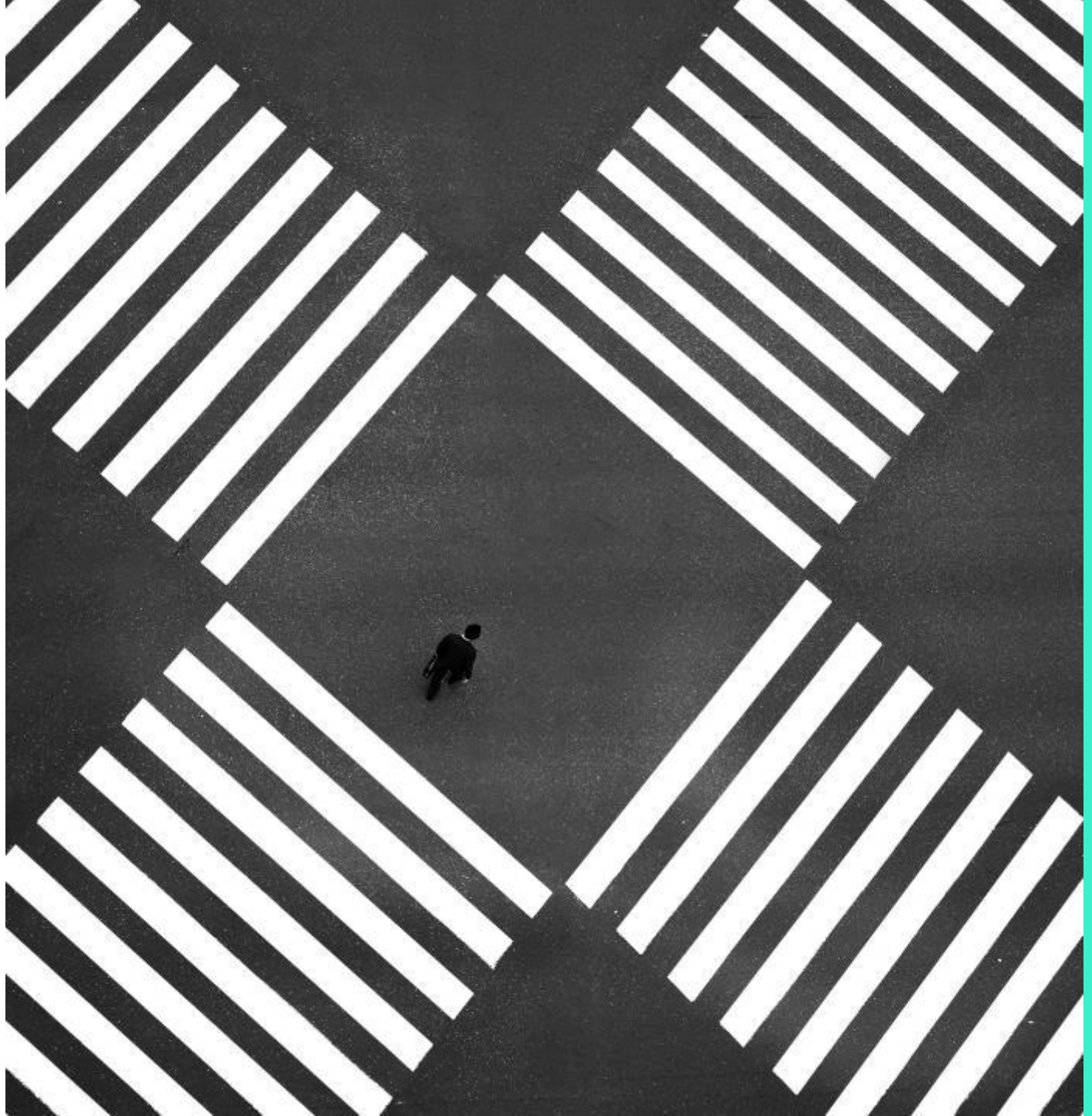
KNN (k=20 optimal): Neighbor-based; AUC=0.904, Accuracy=81.8%–best overal

Selection: KNN superior (higher AUC); confusion matrix: 444 TP (Not Diabetic), 158 TN (Diabetic)..

MODELING: CLEANING IMPACT

Original LR (with Zeros): AUC=0.846, Accuracy=77.2%—underperforms due to biased coeffs (e.g., zeros deflate Glucose effect)

Comparison: Cleaned gains 1.4% accuracy, 0.018 AUC—demonstrates anomalies inflate false negatives.

Insight: Cleaning critical for healthcare reliability; raw data risks missing 5-10% diabetics.

THANK YOU